Floquet Theory applied to Lindbladian Open Quantum Systems

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Floquet Theory

	Closed	Open Lindbladian
State		
Evolution		
Propagator		
Floquet		

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Floquet	$H(t+T) = H(t)$ $U(t,t_0) = P(t,t_0)e^{-\frac{i}{\hbar}(t-t_0)H_{\text{eff}}}$ H_{eff} is Hermitian	$\mathcal{L}(t+T) = \mathcal{L}(t)$ $\mathcal{E}(t,t_0) = \mathcal{P}(t,t_0)e^{(t-t_0)\mathcal{G}_{\text{eff}}}$ $\mathcal{G}_{\text{eff}}$ is ???

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Time Indp. $\mathcal{L}$ CPTP

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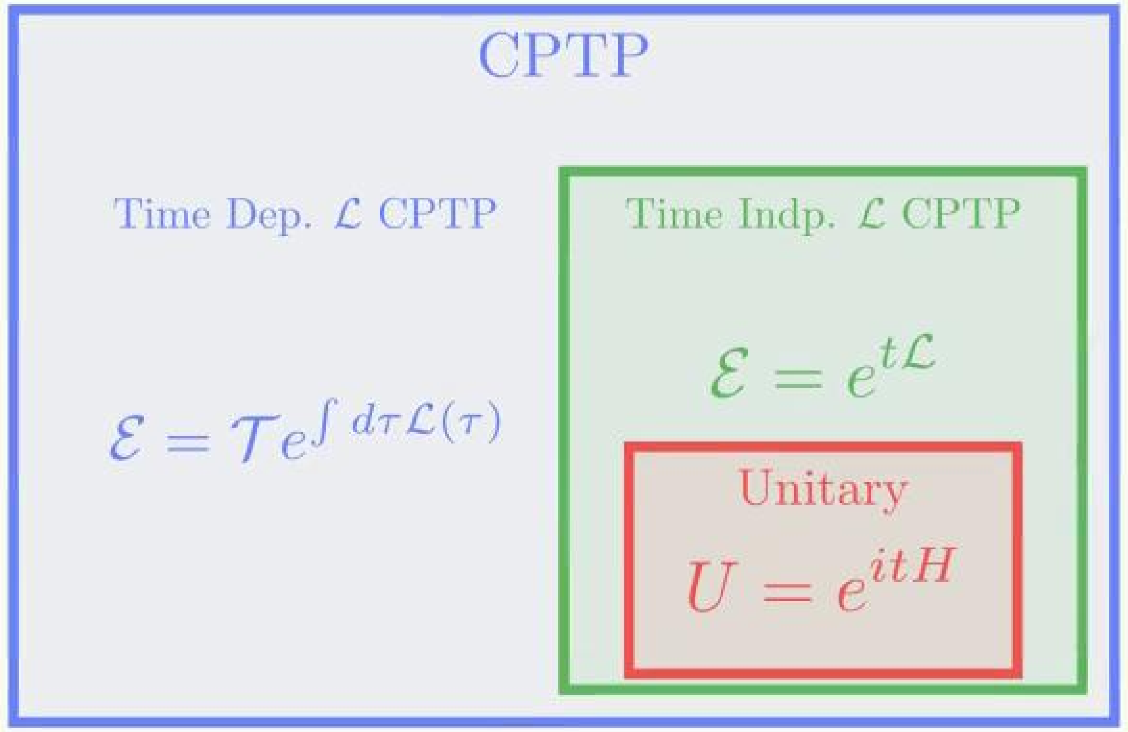

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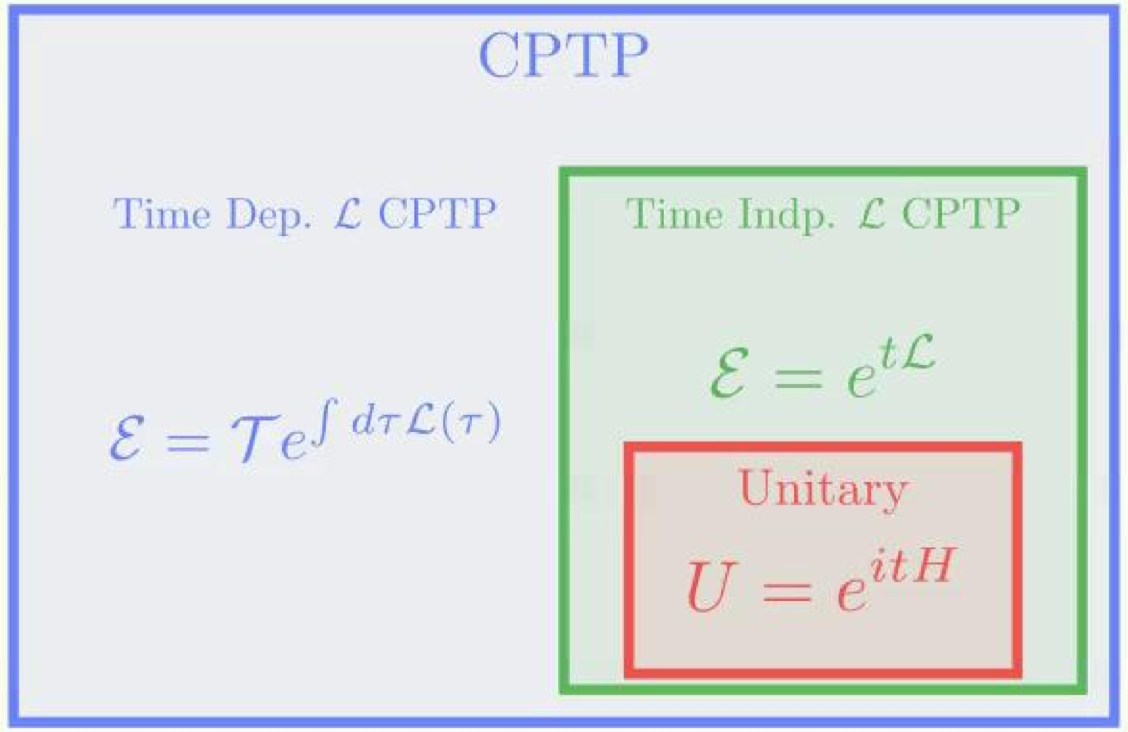

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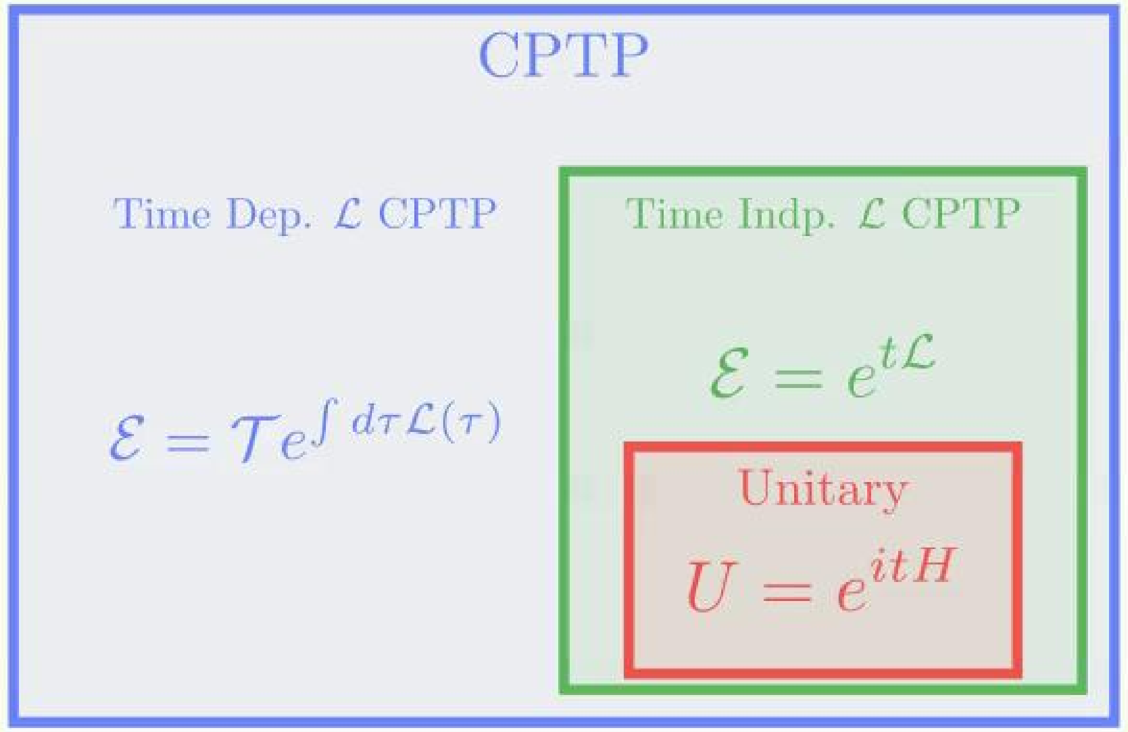

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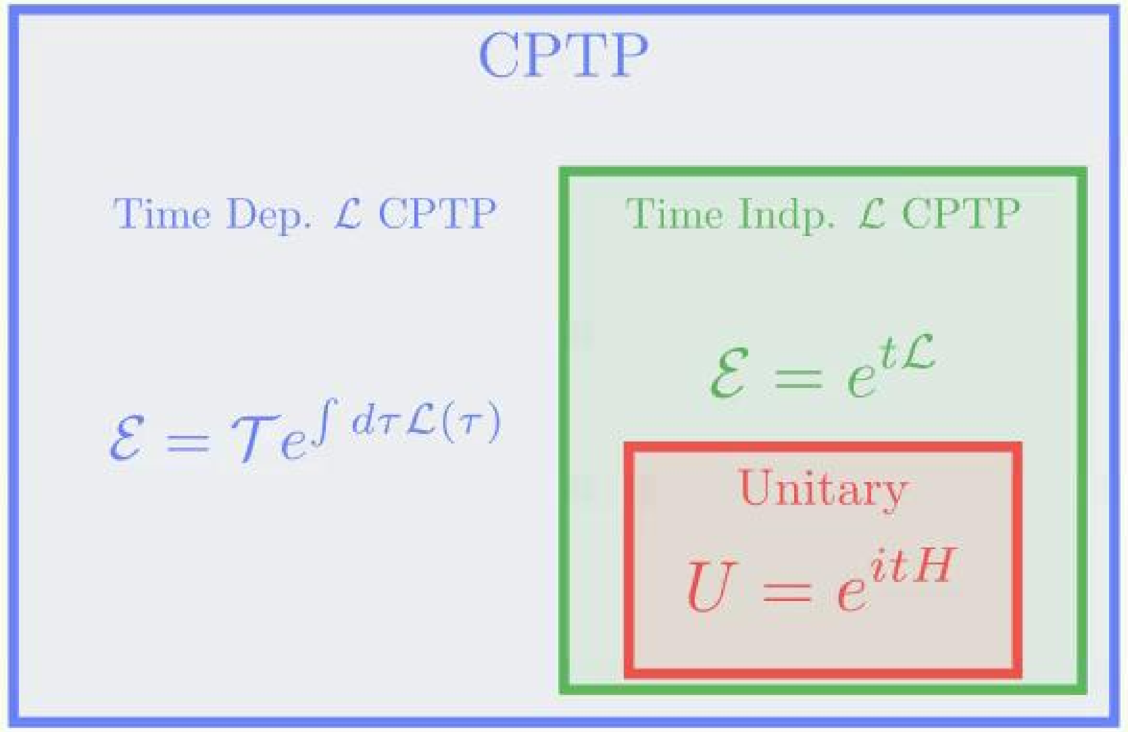

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Only valid for Diagonalizable cases $\rightarrow$ Generalize it. Symmetries

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It is useful to extend this to **Lindbladian** systems

Thank you for your attention!
Any questions?

[**Wolf et al. 2008**]	Wolf, M., Eisert, J., Cubitt, T., & Cirac, J. (2008). *Assessing Non-Markovian Quantum Dynamics* Phys. Rev. Lett., 101, 150402.
[**Schnell et al. 2020**]	Schnell, A., Eckardt, A., & Denisov, S. (2020). *Is there a Floquet Lindbladian?* Phys. Rev. B, 101, 100301.